UNIVERSITY OF ZIMBABWE HMTH111

BSc Honours in Mathematics Level 1

CALCULUS 2

November 2017
Time : 2 hours

Candidates may attempt ALL questions in Section A and at most TWO questions in Section B. Each question should start on a fresh page.

SECTION A (40 marks)

Candidates may attempt ALL questions being careful to number them A1 to A4.

A1. Define what is meant by saying that

(a) a function $f(x, y)$ has a **limit**, L, at a point (a, b) as (x, y) approaches (a, b). [2]

(b) a function $f(x, y)$ has a **partial derivative** $f_x(a, b)$ with respect to x at a point (a, b). [2]

A2. (a) Find the area of the parallelogram determined by $\mathbf{a} = 3\mathbf{i}+4\mathbf{j}$ and $\mathbf{b} = \mathbf{i}+\mathbf{j}+\mathbf{k}$. [5]

(b) Find the volume of a parallelepiped with sides $\mathbf{a} = 3\mathbf{i} - \mathbf{j}$, $\mathbf{b} = \mathbf{j} + 2\mathbf{k}$ and $\mathbf{c} = \mathbf{i} + 5\mathbf{j} + 4\mathbf{k}$. [6]

A3. Let

$$f(x, y) = \begin{cases} \dfrac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0). \end{cases}$$

(a) Determine whether or not $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$ exist at the origin, and find their values if they exist. [6]

(b) Does $\displaystyle\lim_{(x,y)\to(0,0)} f(x, y)$ exist? [6]

A4. (a) Determine whether or not the series converges and find its sum if it converges

$$\sum_{r=1}^{\infty} \frac{1}{(5r-2)(5r+3)} = \frac{1}{3\cdot 8} + \frac{1}{8\cdot 13} + \frac{1}{13\cdot 18} + \cdots + \frac{1}{(5n-2)(5n+3)} + \cdots .$$

[8]

(b) By interchanging the order of integration, or otherwise, evaluate the integral

$$\int_0^1 \int_{3y}^3 e^{x^2}\, dxdy.$$

[5]

SECTION B (60 marks)

Candidates may attempt TWO questions being careful to number them B5 to B8.

B5. (a) (i) Write down the equation of the tangent plane to the paraboloid $z = x^2 + y^2$ at the point $(2, -1, 5)$. [5]

(ii) Find the equation of the plane passing through the points $(2, 4, -3)$, $(3, 7, -1)$ and $(4, 3, 0)$. [5]

(b) (i) Find the angle between the vectors $\mathbf{a} = 2\mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$ and $\mathbf{b} = 5\mathbf{i} + 3\mathbf{j} - 7\mathbf{k}$. [4]

(ii) Find the parametric equations of the line of intersection of the planes

$$2x + y + z = 4 \quad \text{and} \quad 3x - y + z = 3$$

and calculate the angle between them. [8]

(c) (i) Find the parametric and symmetric equations for the straight line through $(2, 5, -7)$ and $(4, 3, 8)$. [4]

(ii) Find the distance between the origin and the plane $x + y + z = 10$. [4]

B6. (a) Using the definition of a limit, show that $\lim\limits_{(x,y)\to(1,2)} x^2 + 2y = 5$. [6]

(b) Find and classify the stationary points of the function

$$f(x, y) = 3x - x^3 - 3xy^2.$$

[6]

(c) Let $w = f(x, y)$, where x and y are given in polar coordinates by the equations $x = r\cos\theta$ and $y = r\sin\theta$.

(i) Obtain expressions for $\dfrac{\partial w}{\partial r}$ and $\dfrac{\partial w}{\partial \theta}$. [8]

(ii) Show that

$$\frac{\partial^2 w}{\partial r^2} = \frac{\partial^2 w}{\partial x^2}\cos^2\theta + 2\frac{\partial^2 w}{\partial x \partial y}\cos\theta\sin\theta + \frac{\partial^2 w}{\partial y^2}\sin^2\theta.$$

[10]

B7. (a) Sketch the region of integration and evaluate the following integrals.

(i) $\displaystyle\int_0^1 \int_x^1 e^{-y^2}\, dydx.$

(ii) $\displaystyle\iint\limits_D \sqrt{x^2+y^2}\, dydx$ where D is the region bounded by $x^2+y^2 \geq 4$ and $x^2+y^2 \leq 9$. [3,3]

(b) (i) Evaluate $\displaystyle\int_1^e \int_0^y \int_0^{\frac{1}{y}} x^2 z\, dxdzdy.$ [5]

(ii) Evaluate the following integral, using an appropriate transformation or otherwise

$$\int_0^1 \int_0^{\sqrt{1-y^2}} \frac{dxdy}{1+x^2+y^2}.$$

[5]

(c) Let $u = x^2 - y^2$ and $v = 2xy$.

(i) Show that $(x^2+y^2)^2 = u^2 + v^2$. [3]

(ii) Hence, find $\dfrac{\partial(x,y)}{\partial(u,v)}$ in terms of u and v. [5]

(iii) Hence, evaluate $\displaystyle\iint\limits_R (x^2+y^2)\, dxdy$, where R is the region in the first quadrant bounded by the hyperbolas $x^2-y^2=6$, $x^2-y^2=1$, $2xy=4$ and $2xy=1$. [6]

B8. (a) Test for convergence or divergence for each of the following infinite series by applying an appropriate test.

(i) $\displaystyle\sum_{n=1}^{\infty} \frac{3^n}{n^2}$ (ii) $\displaystyle\sum_{n=1}^{\infty} \frac{\cos n}{n^2}$ (iii) $\displaystyle\sum_{n=1}^{\infty} \frac{n}{n^2+1}.$ [3,3,3]

(b) (i) Find the sum of the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)}.$ [4]

(ii) Show that the series $\displaystyle\sum_{n=1}^{\infty} \frac{r^n}{n!}$ and $\displaystyle\sum_{n=1}^{\infty} \frac{r^n}{n^n}$ are convergent for all r and that $\displaystyle\sum_{n=1}^{\infty} n!r^n$ and $\displaystyle\sum_{n=1}^{\infty} n^n r^n$ are convergent for no r except for $r=0$. [5]

(c) Test the following series for absolute and conditional convergence and determine the interval of convergence.

(i) $\displaystyle\sum_{n=0}^{\infty} \frac{2^n x^n}{n!}$ (ii) $\displaystyle\sum_{n=1}^{\infty} \frac{x^n}{n3^n}.$ [6,6]

END OF QUESTION PAPER